

TreadmillReviews - Fixed Translation Implementation

Issues Resolved

1. Hydration Error Fixed

- **Problem:** Text content mismatch between server and client (`"TreadmillReviews"` vs `"LaufbandTests"`)
- **Root Cause:** The `LanguageContext` was using `useEffect` to detect locale from URL after hydration, causing server/client content mismatch
- **Solution:** Replaced `useEffect` with `usePathname()` and `useMemo()` to determine locale immediately during render, ensuring server and client render the same content

2. Duplicate Navigation Fixed

- **Problem:** Two navbars and footers appearing when switching languages
- **Root Cause:** Nested layouts - both root `layout.js` and `/de/layout.js` were rendering Header/Footer components
- **Solution:**
 - Removed duplicate `/de/layout.js` file
 - Created single `LayoutWrapper` component that handles dynamic locale and metadata
 - Unified layout structure with proper locale detection

3. Translation Support Across All Routes

- **Problem:** Inconsistent translation support across different pages
- **Solution:** Enhanced `LanguageContext` with proper locale detection that works consistently across all routes

Technical Implementation

Architecture Changes

1. Single Layout System

```
RootLayout -> LayoutWrapper -> LanguageProvider -> Header + Main + Footer
```

2. Hydration-Safe Locale Detection

```
```javascript
// Before (caused hydration errors)
useEffect(() => {
 const path = window.location.pathname;
 if (path.startsWith('/de')) {
 setLocale('de');
 }
}, []);

// After (hydration-safe)
const currentLocale = useMemo(() => {
 if (pathname?.startsWith('/de')) {
 return 'de';
 }
 return 'en';
}, [pathname]);
```

```
}, [pathname]);
``,`
```

## 1. Dynamic Metadata and Lang Attributes

```
```javascript
```

```
// LayoutWrapper handles dynamic html lang and meta tags
```

```
``` ### File Structure
```

```
src/
├─ app/
│ ├─ layout.js (Single root layout)
│ ├─ page.js (English homepage)
│ └─ de/
│ ├─ page.js (German homepage - NO layout.js)
│ ├─ about/page.js
│ └─ blog/page.js
│ ...
├─ components/
│ ├─ LayoutWrapper.js (NEW - handles html/body with dynamic
lang)
│ ├─ Header.js
│ └─ Footer.js
│ ...
└─ contexts/
 └─ LanguageContext.js (FIXED - no more useEffect)
```

## Key Features ###  **\*\*Hydration Error Eliminated\*\*** - Server and client render identical content - No more `useEffect` changing content after hydration - Immediate locale detection from pathname ###  **\*\*Single Navigation System\*\*** - Only one Header and Footer across all pages - No layout duplication - Consistent navigation experience ###  **\*\*Proper Language Switching\*\*** - Works with static export (`output: 'export'`) - Maintains URL structure (`/` vs `/de/`) - Preserves navigation state ###  **\*\*SEO Optimized\*\*** - Dynamic `lang` attribute on `<html>` tag - Locale-specific meta titles and descriptions - Proper OpenGraph tags per language ## Usage Instructions ### Development

```
npm install
npm run dev
```

### ### Production Build

```
npm run build
npm start
```

### Language Switching - English: `http://localhost:3000/` - German: `http://localhost:3000/de/` - Toggle button automatically switches between locales ## Translation Files - English: `messages/en.json` - German: `messages/de.json` ## Browser Support - Works with all modern browsers - No hydration errors in React 18+ - Compatible with Next.js 14+ App Router ## Deployment - Fully compatible with static export - Works on Netlify, Vercel, GitHub Pages - No server-side dependencies required --- ## Migration Notes If you want to migrate from the old implementation: 1. **Remove duplicate layouts**: Delete any nested layout files that include Header/Footer 2. **Update locale detection**: Replace `useEffect` with immediate pathname-based detection 3. **Use single LanguageProvider**: Wrap only at the root level, not in nested layouts 4. **Test thoroughly**: Verify no hydration errors in browser dev tools This implementation provides a robust, hydration-error-free internationalization solution that works perfectly with Next.js static exports.